

09L1-cnn

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Slide 1

Content

CS 24300 - AI Basics

Tony Bergstrom

Slides adapted from Yexiang Xue

Description

This slide is a title slide for a lecture on Artificial Intelligence Basics, likely part of a course called CS 24300. It displays the course title, the instructor's name (Tony Bergstrom), and a credit line indicating that the slides are adapted from another source (Yexiang Xue). The background is a dark gray, and the text is light gray, making it easily readable. The overall presentation is simple and professional, typical of a lecture slide layout.

Figures

Slide 2

Content

Convolutional Neural Network for Image/Video Recognition

Description

This slide, likely part of a lecture on Artificial Intelligence Basics, introduces the topic of Convolutional Neural Networks (CNNs) for image and video recognition. The title is displayed prominently. The slide likely contains images illustrating the application of CNNs to various tasks. The images shown are examples of image recognition tasks, including: (1) face recognition (multiple views of a face with labeled points), (2) autonomous driving (lane detection and vehicle recognition), (3) object detection (detecting objects like a TV, cat, and fruit in a scene), and (4) potentially a digit recognition task (a grid of handwritten digits). The context of the course is focused on the use of CNNs as a powerful technique in image and video analysis within the broader field of AI.

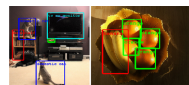
Figures



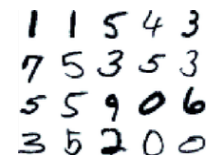
(a)



(b)



(c)



(d)

Slide 3

Content

ImageNet Top-5 Classification Accuracy

15 million images 1000 classes in the ImageNet challenge

[xlabel=Year, ylabel=Classification Error, xmin=2010, xmax=2017, ymin=0, ymax=0.3, xtick=2010,2011,2012,2013,2014,2015,2016,2017, ytick=0,0.05,0.1,0.15,0.2,0.25,0.3, title=ImageNet Top-5 Classification Accuracy, legend pos=north east,] [color=red, fill=red, area legend] coordinates (2010,0.28) (2011,0.28) (2012,0.26) (2013,0.16) (2014,0.12) (2015,0.07) (2016,0.036) (2017,0.023); [color=orange, fill=orange, area legend] coordinates (2010,0.16) (2011,0.16) (2012,0.16) (2013,0.12) (2014,0.07) (2015,0.036) (2016,0.03) (2017,0.023); [color=lightgray, fill=lightgray, area legend] coordinates (2010,0.12) (2011,0.12) (2012,0.12) (2013,0.12) (2014,0.07) (2015,0.036) (2016,0.03) (2017,0.023); [color=blue, fill=blue, area legend] coordinates (2010,0.07) (2011,0.07) (2012,0.07) (2013,0.07) (2014,0.07) (2015,0.036) (2016,0.03) (2017,0.023); [above right] at (axis cs:2012,0.26) AlexNet, The Beginning; [above right] at (axis cs:2015,0.07) The first* fast** GPU-accelerated Deep Convolutional; [above right] at (axis cs:2015,0.05) Neural Network to win an image recognition contest; [above right] at (axis cs:2015,0.07) 16.7% ↓ 23.3% ↓;